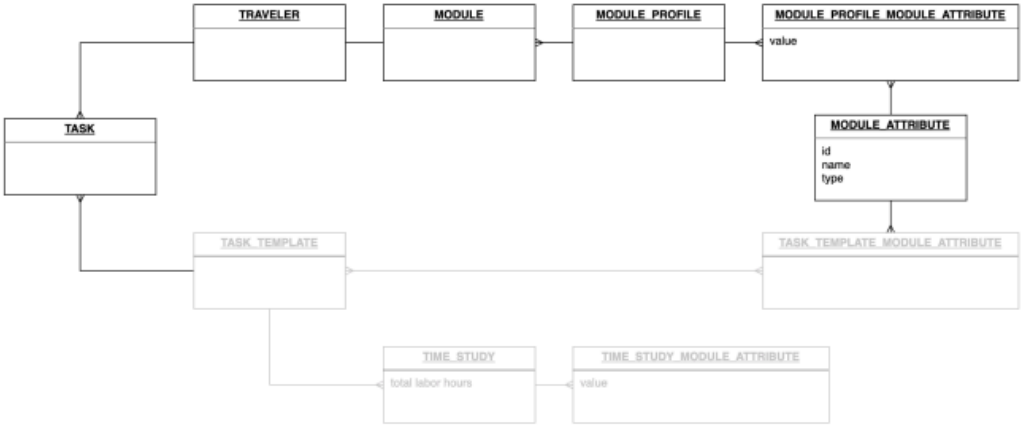
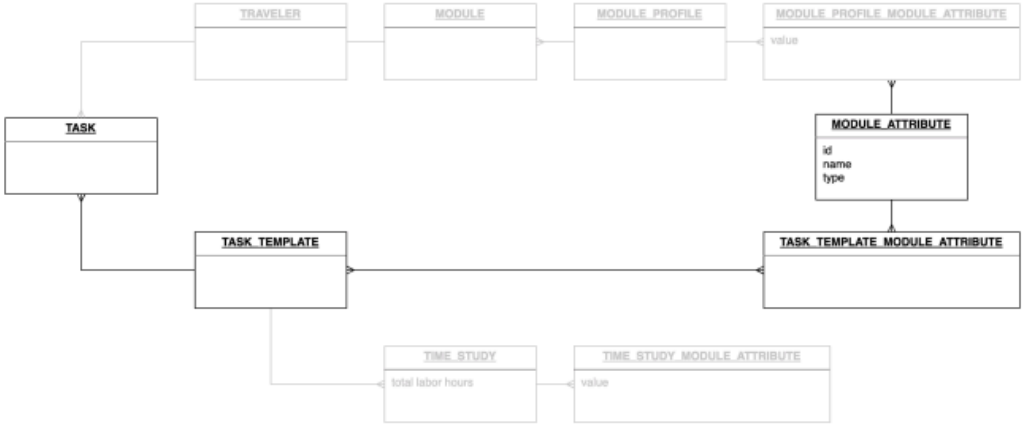
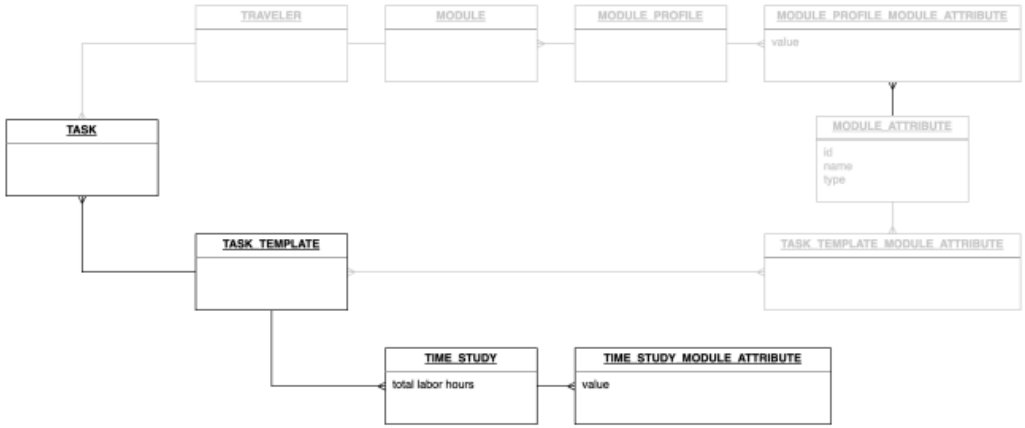


A task is performed on a production unit, which we call a module. A module has certain attributes like linear feet of interior walls or number of toilets or square footage of roof. The first step to estimate the time for a task is to get all of the attributes of the module. To do this, retrieve all of the MODULE_PROFILE_MODULE_ATTRIBUTE records. These have a value and an identifier for the specific attribute that value represents.



The next step is to see which attributes apply or are relevant to this task. To do this, retrieve all the TASK_TEMPLATE_MODULE_ATTRIBUTE records. These do not have a value. They exist to define, per task template (and task), which attributes apply to that task. Then, apply the list of TASK_TEMPLATE_MODULE_ATTRIBUTE to the list of MODULE_PROFILE_MODULE_ATTRIBUTE to get an intersection. That is, the list of. module profile attributes whose ID is among the task template attributes. The found module profile attributes and their `value` define the work that is to be done and is compared against the 'reference' value, which is needed next. For example, if the the task is a drywall task, then the attributes and values found would be linear feet of interior walls and maybe count of interior walls.



The next step is to compare the work to be done to a time study and calculate our estimate. Get the TIME_STUDY_MODULE_ATTRIBUTES records for the time study. You should get the same count and same attributes here as the MODULE_PROFILE_MODULE_ATTRIBUTE found in the previous step. If there are time study attributes that are not represented in the module profile attributes found, throw an error. The time study has total labor hours spent for the work performed. For example, the time study will show that performing a drywall task for 92 linear feet of indoor walls and a count of 6 indoor walls took 18 hours. Now compare this to the work that is to be performed. Let's say, 72 linear feet and 8 walls. Use simple proportions with equal weighting for each attribute. In this example, we have a -22% reduction in the linear feet of walls and a 33% increase in the wall count resulting in a net change of +11%. Which gives an estimate of 18 hours from the time study + 11% equals 19.8 hours.